

Q1 - 24 June - Shift 1

A parallel plate capacitor is formed by two plates each of area $30\pi \text{ cm}^2$ separated by 1 mm. A material of dielectric strength $3.6 \times 10^7 \text{ Vm}^{-1}$ is filled between the plates. If the maximum charge that can be stored on the capacitor without causing any dielectric breakdown is $7 \times 10^{-6} \text{ C}$, the value of dielectric constant of the material is :

$$\left\{ \text{Use : } \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2\text{C}^{-2} \right\}$$

- (A) 1.66 (B) 1.75
(C) 2.25 (D) 2.33

Space for your notes:

Q2 - 24 June - Shift 2

If the charge on a capacitor is increased by 2 C, the energy stored in it increases by 44%. The original charge on the capacitor is (in C) :

- (A) 10 (B) 20
(C) 30 (D) 40

Space for your notes:

Q3 - 25 June - Shift 1

The equivalent capacitance between points A and B in below shown figure will be _____ μF .

Space for your notes:



Q4 - 25 June - Shift 2

Questions

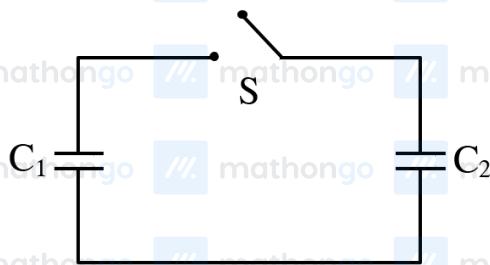
MathonGo

Two metallic plates form a parallel plate capacitor. The distance between the plates is 'd'. A metal sheet of thickness $\frac{d}{2}$ and of area equal to area of each plate is introduced between the plates. What will be the ratio of the new capacitance to the original capacitance of the capacitor?

- (A) 2:1 (B) 1:2
(C) 1:4 (D) 4:1

*Space for your notes:***Q5 - 26 June - Shift 1**

Two capacitors having capacitance C_1 and C_2 respectively are connected as shown in figure. Initially, capacitor C_1 is charged to a potential difference V volt by a battery. The battery is then removed and the charged capacitor C_1 is now connected to uncharged capacitor C_2 by closing the switch S . The amount of charge on the capacitor C_2 , after equilibrium is :

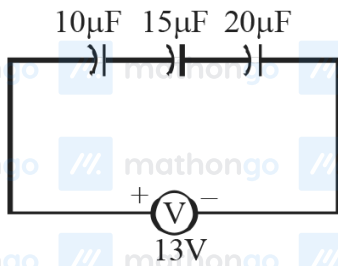
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- (A) $\frac{C_1 C_2}{(C_1 + C_2)} V$ (B) $\frac{(C_1 + C_2)}{C_1 C_2} V$
(C) $(C_1 + C_2) V$ (D) $(C_1 - C_2) V$

Q6 - 26 June - Shift 2

#MathBoleTohMathonGo

The charge on capacitor of capacitance $15\mu\text{F}$ in the figure given below is :



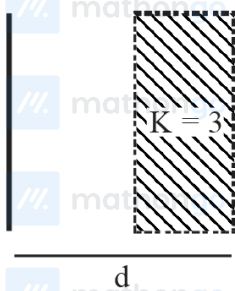
- (A) $60\mu\text{C}$ (B) $130\mu\text{C}$ (C) $260\mu\text{C}$ (D) $585\mu\text{C}$

Space for your notes:

Q7 - 26 June - Shift 2

A parallel plate capacitor with plate area A and plate separation $d=2\text{ m}$ has a capacitance of $4\mu\text{F}$.

The new capacitance of the system if half of the space between them is filled with a dielectric material of dielectric constant $K=3$ (as shown in figure) will be :



- (A) $2\mu\text{F}$ (B) $32\mu\text{F}$ (C) $6\mu\text{F}$ (D) $8\mu\text{F}$

Space for your notes:

Q8 - 27 June - Shift 1

Questions

MathonGo

A force of 10N acts on a charged particle placed between two plates of a charged capacitor. If one plate of capacitor is removed, then the force acting on that particle will be :

- (A) 5 N (B) 10 N
(C) 20 N (D) Zero

*Space for your notes:***Q9 - 27 June - Shift 1**

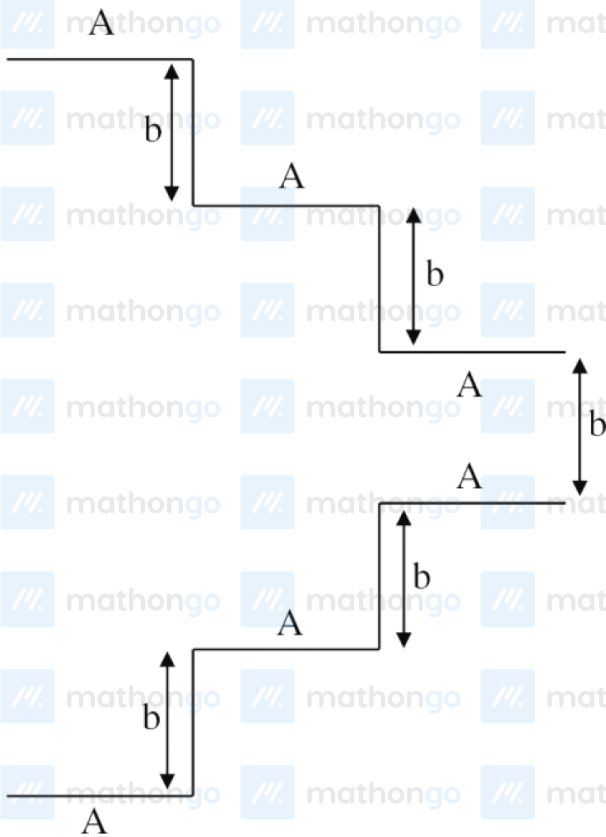
A capacitor of capacitance 50 pF is charged by 100 V source. It is then connected to another uncharged identical capacitor. Electrostatic energy loss in the process is _____ nJ.

*Space for your notes:***Q10 - 27 June - Shift 2**

#MathBoleTohMathonGo

A parallel plate capacitor is made up of stair like structure with a plate area A of each stair and that is connected with a wire of length b , as shown in the figure. The capacitance of the arrangement is

$\frac{x}{15} \frac{\epsilon_0 A}{b}$. The value of x is _____.



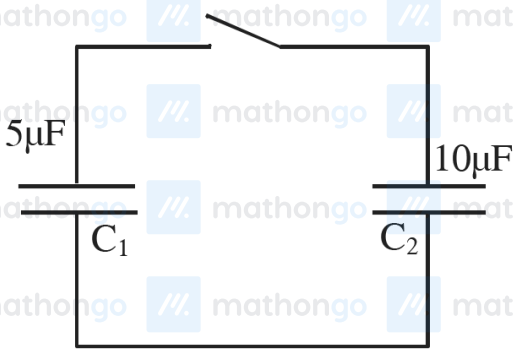
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Q11 - 28 June - Shift 2

Questions

MathonGo

A capacitor C_1 of capacitance $5\mu\text{F}$ is charged to a potential of 30 V using a battery. The battery is then removed and the charged capacitor is connected to an uncharged capacitor C_2 of capacitance $10\mu\text{F}$ as shown in figure. When the switch is closed charge flows between the capacitors. At equilibrium, the charge on the capacitor C_2 is _____ μC .

*Space for your notes:***Q12 - 29 June - Shift 1**

A parallel plate capacitor filled with a medium of dielectric constant 10, is connected across a battery and is charged. The dielectric slab is replaced by another slab of dielectric constant 15. Then the energy of capacitor will :

- (A) increase by 50% (B) decrease by 15%
(C) increase by 25% (D) increase by 33%

*Space for your notes:***Q13 - 29 June - Shift 2**

#MathBoleTohMathonGo

Questions

MathonGo

A capacitor is discharging through a resistor R.

Consider in time t_1 , the energy stored in the capacitor reduces to half of its initial value and in time t_2 , the charge stored reduces to one eighth of its initial value. The ratio t_1/t_2 will be :

- (A) $1/2$ (B) $1/3$
(C) $1/4$ (D) $1/6$

Space for your notes:

Q14 - 29 June - Shift 2

The displacement current of $4.425 \mu\text{A}$ is developed in the space between the plates of parallel plate capacitor when voltage is changing at a rate of 10^6 Vs^{-1} . The area of each plate of the capacitor is 40 cm^2 . The distance between each plate of the capacitor is $x \times 10^{-3} \text{ m}$. The value of x is,

(Permittivity of free space, $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$)

Space for your notes:

#MathBoleTohMathonGo

Answer Key

Q1 (D)	Q2 (A)	Q3 (6)	Q4 (A)
Q5 (A)	Q6 (A)	Q7 (C)	Q8 (A)
Q9 (125)	Q10 (23)	Q11 (100)	Q12 (A)
Q13 (D)	Q14 (8)		

#MathBoleTohMathonGo

Q1 (D)

$$K = \frac{q}{A \epsilon_0 E} = \frac{7 \times 10^{-6}}{30\pi \times 10^{-4} \times \frac{1}{4\pi \times 9 \times 10^9} \times 3.6 \times 10^7}$$

$$K = \frac{36 \times 7}{30 \times 3.6} = 2.33$$

Q2 (A)

$$U \propto q^2$$

$$\Rightarrow q_f = 1.2 q$$

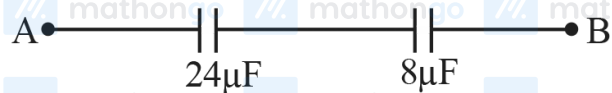
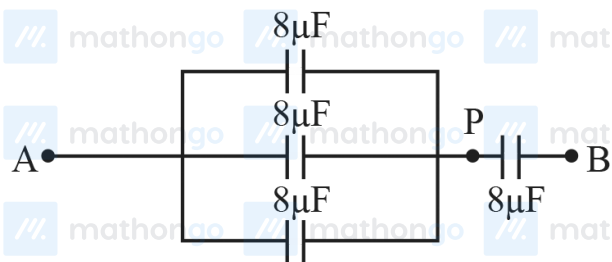
$$q_f - q = 2$$

$$\Rightarrow 1.2 q - q = 2$$

$$q = 10$$

Q3 (6)

Two capacitors are short circuited



Finally equivalent capacitance

$$= \frac{24 \times 8}{24 + 8} = \frac{24 \times 8}{32} = 6\mu\text{F}$$

Q4 (A)

$$C_1 = \frac{\epsilon_0 A}{d}$$

$$C_2 = \frac{\epsilon_0 A}{\frac{d}{2} + \frac{d/2}{\infty}} = \frac{2\epsilon_0 A}{d}$$

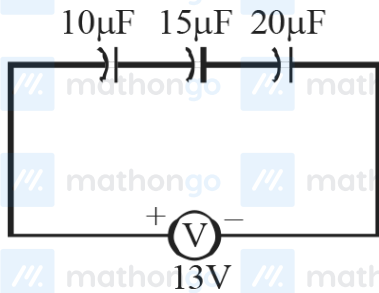
$$\frac{C_2}{C_1} = \frac{2}{1}$$

Q5 (A)

Charge on capacitor C_2

$$= \frac{C_2 \times Q_{\text{total}}}{C_{\text{total}}} = \frac{C_2 [C_1 V]}{C_1 + C_2} = \frac{C_1 C_2 V}{C_1 + C_2}$$

Q6 (A)



$$\frac{1}{C_{\text{eq}}} = \frac{1}{10} + \frac{1}{15} + \frac{1}{20} = \frac{12 + 8 + 6}{120} = \frac{26}{120}$$

$$C_{\text{eq}} = \frac{60}{13} \mu\text{F}$$

$$Q = \frac{13 \times 60}{13} = 60 \mu\text{C}$$

Charge on each capacitor is same

$\therefore$ they are in series.

Q7 (C)

$$C_{\text{original}} = \frac{A\epsilon_0}{d}$$



$$C_1 = \frac{A\epsilon_0}{d/2} = \frac{2A\epsilon_0}{d} = C$$

$$C_2 = \frac{KA\epsilon_0}{d/2} = \frac{2KA\epsilon_0}{d} = \frac{6A\epsilon_0}{d} = 3C$$

C_1 & C_2 are in series

$$C_{\text{new}} = \frac{C_1 C_2}{C_1 + C_2} = \frac{C \times 3C}{C + 3C} = \frac{3C}{4}$$

$$= \frac{3}{4} \times \frac{2A\epsilon_0}{d} = \frac{3}{2} \times \frac{A\epsilon_0}{d}$$

$$C_{\text{new}} = \frac{3}{2} C_{\text{original}}$$

$$= \frac{3}{2} \times 4 = 6\mu\text{F}$$

Q8 (A)

$$X = A \sin \omega t \left(t = 3, X = \frac{A}{2} \right)$$

$$\Rightarrow \frac{A}{2} = A \sin 3\omega$$

$$\Rightarrow \sin 3\omega = \frac{1}{2}$$

$$\Rightarrow 3\omega = \frac{\pi}{6}$$

$$\Rightarrow \omega = \frac{\pi}{18} = \frac{2\pi}{T}$$

$$\Rightarrow T = 36s$$

Q9 (125)

$$\text{Energy loss} = \frac{1}{2} \frac{C_1 C_2}{C_1 + C_2} (V_1 - V_2)^2$$

$$= \frac{1}{2} \frac{50 \times 50 \times 10^{-12} \times 10^{-12}}{(50 + 50) 10^{-12}} (100 - 0)^2 = 125 nJ$$

Q10 (23)

Parallel combination

$$c_{eq} = \epsilon_0 A \left[\frac{1}{5b} + \frac{1}{3b} + \frac{1}{b} \right] = \frac{23}{15} \frac{\epsilon_0 A}{b}$$

Q11 (100)

Before closing the switch

$$Q = C_1 V_0 = 5 \times 30 = 150 \mu\text{C}$$

After closing the switch

$$V = \frac{Q}{C_1 + C_2} = \frac{150}{10 + 5} = 10 \text{ V}$$

$$Q_2 = C_2 V = 10 \times 10 = 100 \mu\text{C}$$

Q12 (A)

$$E \Rightarrow \frac{1}{2}(KC)V^2$$

$\therefore$ % change

$$\Rightarrow \frac{\frac{1}{2}K_2 CV^2 - \frac{1}{2}K_1 CV^2}{\frac{1}{2}K_1 CV^2} = \frac{K_2 - K_1}{K_1} \times 100$$

$$\Rightarrow \frac{15 - 10}{10} \times 100 = 50\%$$

Q13 (D)

In t_1 time energy becomes half so charge will

become $\frac{1}{\sqrt{2}}$ time

$$q = Q_0 e^{-\frac{t_1}{RC}} = \frac{Q_0}{\sqrt{2}}$$

$$\text{and } q = Q_0 e^{-\frac{t_1}{RC}} = \frac{Q_0}{8} = \left(\frac{Q_0}{\sqrt{2}}\right)^6$$

$$t_2 = 6t_1$$

$$\frac{t_1}{t_2} = \frac{1}{6}$$

Q14 (8)

Displacement Current = Conduction Current

$$= \frac{dq}{dt}$$

$$I_d = \frac{\epsilon_0 A}{d} \frac{dV}{dt}$$

$$d = \frac{8.85 \times 10^{-12} \times 4 \times 10^{-3} \times 10^6}{4.425 \times 10^{-6}}$$

$$= 8 \text{ mm}$$

$$X = 8$$